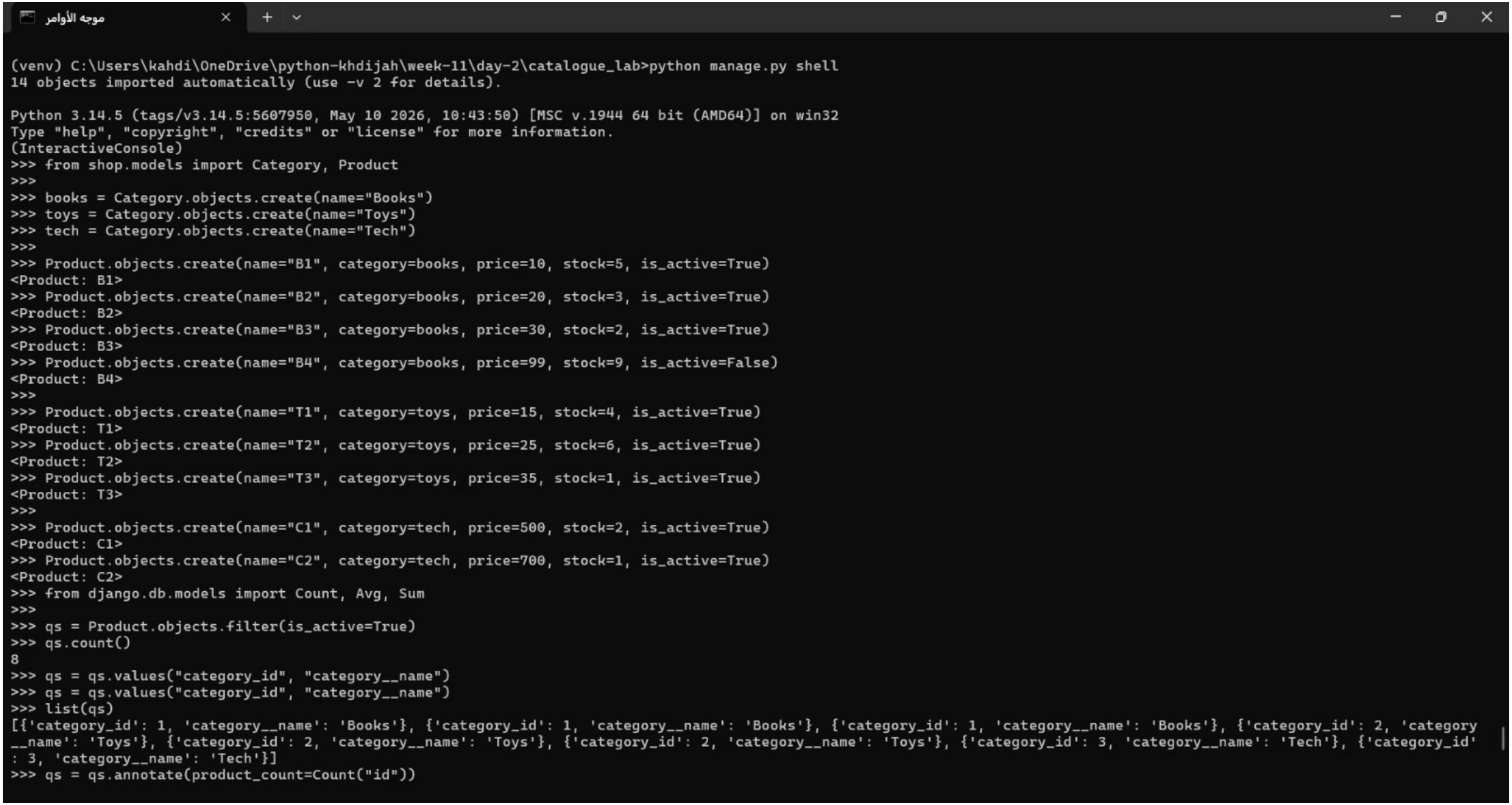

موجه الأوامر
(venv) C:\Users\kahdi\OneDrive\python-khdijah\week-11\day-2\catalogue_lab>python manage.py shell
14 objects imported automatically (use -v 2 for details).
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
(InteractiveConsole)
>>> from shop.models import Category, Product
>>>
>>> books = Category.objects.create(name="Books")
>>> toys = Category.objects.create(name="Toys")
>>> tech = Category.objects.create(name="Tech")
>>>
>>> Product.objects.create(name="B1", category=books, price=10, stock=5, is_active=True)
<Product: B1>
>>> Product.objects.create(name="B2", category=books, price=20, stock=3, is_active=True)
<Product: B2>
>>> Product.objects.create(name="B3", category=books, price=30, stock=2, is_active=True)
<Product: B3>
>>> Product.objects.create(name="B4", category=books, price=99, stock=9, is_active=False)
<Product: B4>
>>>
>>> Product.objects.create(name="T1", category=toys, price=15, stock=4, is_active=True)
<Product: T1>
>>> Product.objects.create(name="T2", category=toys, price=25, stock=6, is_active=True)
<Product: T2>
>>> Product.objects.create(name="T3", category=toys, price=35, stock=1, is_active=True)
<Product: T3>
>>>
>>> Product.objects.create(name="C1", category=tech, price=500, stock=2, is_active=True)
<Product: C1>
>>> Product.objects.create(name="C2", category=tech, price=700, stock=1, is_active=True)
<Product: C2>
>>> from django.db.models import Count, Avg, Sum
>>>
>>> qs = Product.objects.filter(is_active=True)
>>> qs.count()
8
>>> qs = qs.values("category_id", "category__name")
>>> qs = qs.values("category_id", "category__name")
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books'}, {'category_id': 1, 'category__name': 'Books'}, {'category_id': 1, 'category__name': 'Books'}, {'category_id': 2, 'category__name': 'Toys'}, {'category_id': 2, 'category__name': 'Toys'}, {'category_id': 2, 'category__name': 'Toys'}, {'category_id': 3, 'category__name': 'Tech'}, {'category_id': 3, 'category__name': 'Tech'}]
>>> qs = qs.annotate(product_count=Count("id"))

```
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books', 'product_count': 3}, {'category_id': 2, 'category__name': 'Toys', 'product_count': 3}, {'category_id': 3, 'category__name':
 'Tech', 'product_count': 2}]
>>> qs = qs.annotate(
...     average_price=Avg("price"),
...     total_stock=Sum("stock"),
... )
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books', 'product_count': 3, 'average_price': Decimal('20'), 'total_stock': 10}, {'category_id': 2, 'category__name': 'Toys', 'produ
ct_count': 3, 'average_price': Decimal('25'), 'total_stock': 11}, {'category_id': 3, 'category__name': 'Tech', 'product_count': 2, 'average_price': Decimal('600'), 'total
_stock': 3}]
>>> qs = qs.filter(product_count__gte=3)
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books', 'product_count': 3, 'average_price': Decimal('20'), 'total_stock': 10}, {'category_id': 2, 'category__name': 'Toys', 'produ
ct_count': 3, 'average_price': Decimal('25'), 'total_stock': 11}]
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books', 'product_count': 3, 'average_price': Decimal('20'), 'total_stock': 10}, {'category_id': 2, 'category__name': 'Toys', 'produ
ct_count': 3, 'average_price': Decimal('25'), 'total_stock': 11}]
>>> qs = qs.order_by("-product_count", "category__name")
>>> list(qs)
[{'category_id': 1, 'category__name': 'Books', 'product_count': 3, 'average_price': Decimal('20'), 'total_stock': 10}, {'category_id': 2, 'category__name': 'Toys', 'produ
ct_count': 3, 'average_price': Decimal('25'), 'total_stock': 11}]
>>> overall_summary = Product.objects.filter(is_active=True).aggregate(
...     total_products=Count("id"),
...     average_price=Avg("price"),
... )
>>>
>>> overall_summary
{'total_products': 8, 'average_price': Decimal('166.875')}
>>>
KeyboardInterrupt
>>>
>>> exit()
now exiting InteractiveConsole...

(venv) C:\Users\kahdi\OneDrive\python-khdijah\week-11\day-2\catalogue_lab>git add .

(venv) C:\Users\kahdi\OneDrive\python-khdijah\week-11\day-2\catalogue_lab>cd ..

(venv) C:\Users\kahdi\OneDrive\python-khdijah\week-11\day-2>git add .

(venv) C:\Users\kahdi\OneDrive\python-khdijah\week-11\day-2>git commit -m "Add"
[master 28a1e64] Add
 25 files changed, 271 insertions(+)
 create mode 100644 week-11/day-2/Explain (1).pdf
```